

- ### SECTION-D
- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x8=16)
- Q.23 Discuss in detail the conventional manufacturing line and latest automated manufacturing. (CO1)
- Q.24 What is solid modeling? Explain hidden line removal algorithm also. (CO3)
- Q.25 Write short notes on (CO5)
- a) Tool Length and Radius Compensation
 - b) Constructive Solid Geometry(CSG)

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- a) CAD b) CAM
- c) CAQ d) CAE

Q.4 In a DNC system (CO4)

- a) many machine tools can be controlled simultaneously
- b) only a single machine tool is controlled
- c) NC machine cannot be controlled
- d) none of the above

Q.5 The DDA algorithm is mainly used for: (CO2)

- a) Filling colors b) Textures
- c) Line generation d) Curve smoothing

Q.6 Which drive system is commonly used in CNC machines? (CO5)

- a) Mechanical b) Hydraulic
- c) Pneumatic d) Both b and c

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 CSG stands for _____. (CO3)

Q.8 The term B-rep stands for _____. (CO3)

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Q.9 The G01 code is for _____ interpolation. (CO4)

Q.10 Define CAM. (CO1)

Q.11 Enlist the type of file formats used in CAD/CAM. (CO1)

Q.12 Which code is used for circular interpolation in CNC programming _____. (CO4)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Explain the scope of CAD/CAM. (CO1)

Q.14 Differentiate between close and open loop systems. (CO4)

Q.15 What are the various advantages of CAM over conventional manufacturing? (CO5)

Q.16 Write the advantages of analytical curves. (CO3)

Q.17 Explain sweep representation in geometry modelling. (CO3)

Q.18 What is geometric modeling? (CO3)

Q.19 Explain the concept of tool length and radius compensation. (CO4)

Q.20 Explain the types of adaptive control in CNC/DNC. (CO5)

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